Exhaust Gas Cleaning Systems (resolution MEPC.340(77))
Performance Level 1

See section 4.1.1 for the requirements for Performance Level 1. This performance level, where limited, is comparable to the existing requirements for incinerators (resolution MEPC.244(66), as amended) but as appropriate to in-service monitoring applied on a

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This performance level, where limited, is comparable to the existing requirements for incinerators (resolution MEPC.244(66), as amended) but as appropriate to in-service monitoring applied on a continuous basis to a thermal waste treatment device
Performance Level 2

Requirements set out in section 4.1.2. This level has tighter emission to air limits than Performance Level 1. The report from that testing should be included in the TWTD Technical Report
Sewage sludge Material from the ship's sewage system which would include, but not be limited to, de -watered sewage prior to treatment or the residues from a sewage treatment plant
Supporting information
annex
Commercially sensitive material submitted to the approving Administration covering detailed aspects of the TWTD Technical Report, which should not be circulated outside that Administration
Thermal waste
treatment device
(TWTD)

A device for disposing, by thermal action, of onboard generated garbage other than by use of an incinerator as defined by paragraph 2.2 of the 2014 Standard specification for shipboard incinerators (resolution MEPC.244(66), as amended).

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A device for disposing, by thermal action, of onboard generated garbage other than by use of an incinerator as defined by paragraph 2.2 of the 2014 Standard specification for shipboard incinerators (resolution MEPC.244(66), as amended).
The thermal waste treatment device includes the waste reduction unit itself together with all other necessary support systems and equipment

TWTD File The document prepared by the applicant for each certified thermal

waste treatment device and approved by the Administration.
The TWTD File should be retained on board with the device during its service life. The TWTD File details the device and how it is to be surveyed or inspected
TWTD Operating Manual
The document supplied with the thermal waste treatment device describing to the user how the device is to be installed, operated, maintained and serviced
TWTD Technical Report
The document prepared by the applicant detailing how the Functional Objectives are met.

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SOURCE: MEPC.373(80).pdf
describing to the user how the device is to be installed, operated, maintained and serviced
TWTD Technical Report
The document prepared by the applicant detailing how the Functional Objectives are met.
The TWTD Technical Report would form part of the information supplied to the Organization by the Administration of a Party approving a thermal waste treatment device as an equivalent means in accordance with the requirements of regulation 4 of MARPOL Annex VI
UTC Universal Time Coordinated

4 Emission limits

4.1 Discharges to air

This section does not apply to systems which do not generate any emissions to air, such as hydrothermal carbonization (HTC).

4.1.1 Performance Level 1

4.1.1.1 A TWTD certified under these Guidelines should not exceed the following in -service maximum emission limits:

CO 185 ppm (dry basis) at 11.00% O2 averaged over each UTC three-hour period

Soot number maximum average: Bacharach 3 or Ringelman 1 (20% opacity) (A higher soot number is acceptable only during very short periods such as starting up)

4.1.1.2 CO should be measured in accordance with section 6.4 of the NO x Technical

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SOURCE: MEPC.373(80).pdf
Soot number maximum average: Bacharach 3 or Ringelman 1 (20% opacity) (A higher soot number is acceptable only during very short periods such as starting up)

4.1.1.2 CO should be measured in accordance with section 6.4 of the NO x Technical Code 2008 (direct measurement and monitoring) and should be monitored at a frequency of not less than 0.05 Hz.

4.1.1.3 Oxygen content, temperature and pressure profiles, as applicable, through the TWTD should be monitored and controlled in accordance with the relevant Functional Objective.

4.1.2 Performance Level 2

4.1.2.1 Where requested by the applicant, the TWTD may additionally be certified as meeting Performance Level 2. This involves a detailed pre -certification test together with in -service emission limits which are tighter than those of Performance Level 1.

4.1.3 Pre-approval Test

4.1.3.1 As part of the initial approval process for Performance Level 2 as an equivalent means, the TWTD model should be subject to a Pre-approval Test.

4.1.3.2 The Pre-approval Test should be of 6 to 8 hours duration with the TWTD in its operating condition.
RESOLUTION MEPC.373(80) (adopted on 7 July 2023)

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SOURCE: MEPC.373(80).pdf
the TWTD model should be subject to a Pre-approval Test.

4.1.3.2 The Pre-approval Test should be of 6 to 8 hours duration with the TWTD in its operating condition.
RESOLUTION MEPC.373(80) (adopted on 7 July 2023)

4.1.3.3 Pre-approval Test emission limit values are given on a dry basis, other than for HC which is measured on a wet basis, at 11.00 % O2 concentration and at 273 K, 101.3 kPa:

Pre-approval Test

Species	Limit	Test Method
CO	50 ppm	NTC **
NOx	100 ppm as NO2	NTC **
HC	15 ppmC1	NTC **
Particulate matter	10 mg/m3	US EPA Method 5
Hydrogen Chloride (HCl)	10 mg/m3	US EPA Method 26/26A
Dioxins and Furans*	0.1 ng/m3	US EPA Method 1613B

In addition oxygen, temperature and pressure profiles, as applicable, through the device should be monitored over the duration of the test period and given in the test report.***

* as listed and with equivalency calculated in accordance with EU Directive 2010/75/EU Annex-VI Part 2

** in accordance with the NOx Technical Code 2008 chapter 5

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SOURCE: MEPC.373(80).pdf
* as listed and with equivalency calculated in accordance with EU Directive 2010/75/EU Annex-VI Part 2
** in accordance with the NOx Technical Code 2008 chapter 5
*** oxygen content, temperature and pressure, as applicable, should be measured in accordance with NOx Technical Code 2008 chapter 5

4.1.3.4 Pre-approval Test procedures:

.1 testing should be undertaken while operating with a) sludge oil (if applicable to the TWTD system) and b) solid waste compositions both as given in paragraph 1 of appendix IV to MARPOL Annex VI;

.2 sampling position should be after any exhaust gas treatment components, such as water washing, but prior to any dilution of that exhaust gas;

.3 CO, NOx and HC should be monitored at a frequency of not less than 0.05 Hz over the duration of each test and those readings averaged to give the result to be compared to the respective limit value;